

Agentic Coding and Generative AI Research Tools

Christopher Stubbs

Department of Physics & Department of Astronomy

Harvard University

stubbs@physics.harvard.edu

Welcome!



What's Generative AI?

- A suite of tools (ChatGPT, Claude, Gemini, etc.) that are trained on an extensive corpus of information (text, images, etc.) and can make a statistical prediction of next-word in a sequence.
- The dog that I saw was _____.

Brown	0.76
Running	0.12
Dead	0.05
Hungry	0.01
- Simple, right? Well.... maybe not. Remarkable capabilities from next-token completion!
- ChatGPT was released Nov 2022. Caught nearly everyone off-guard.
- Explosive growth in capabilities of GAI models, ever since.

*Unless you've tried cutting edge GAI tools in the last month,
your impressions are out of date and irrelevant.*

Non-reproducible output!

- Stochastic aspects make *outputs* not repeatable.
- What you get back depends on
 - History (context window retains a certain lookback history)
 - Prompts you give it
 - “hidden” system prompts
 - Vendor and version of AI system
 - Parameter settings (‘temperature’, etc).

As AI tools migrate further upstream in our workflow, will this non-reproducibility collide with the scientific method?



Use case #1) AI can *accelerate* most of the tasks we do

- Background research and learning new things
- Interactive brainstorming
- Interactive data browsing and knowledge exploration
- Code generation and debugging
- AI-automated workflow
 - Code -> cybersecurity tests -> unit tests -> documentation update -> github update
- Generation of summaries on topics
- Generation of figures
- Circuit board design and 4-layer board routing
- Automated generation of user manuals and documentation
- Transcripts of meetings and talks
- Translation of slides to Japanese before an interview
- Visualizations, both static and dynamic
- Compliance verification for proposals, pre-submission

- Narrative text – outlines, drafts, polishing, reviewing.....



Use case #2) AI is pushing back the frontier of knowledge

nature

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NEWS | 18 September 2026

AI cracked the Navier–Stokes challenge. What does that mean for physics?

Physicists and mathematicians are going beyond the classic equations of fluid dynamics to understand turbulence – often with the help of AI.

Nature of science?
Verification and validation?
Inequity of resource access?

....



HARVARD
Faculty of Arts and Sciences

mathandai.org

✓ Pinger

✓ Space Force tutori...

🗄

📁 LSST commission...

📅 Google Calendar

📰 Google News

☰ AA to do - Google...

Math and AI.

A Severe Misalignment of AI in Mathematics

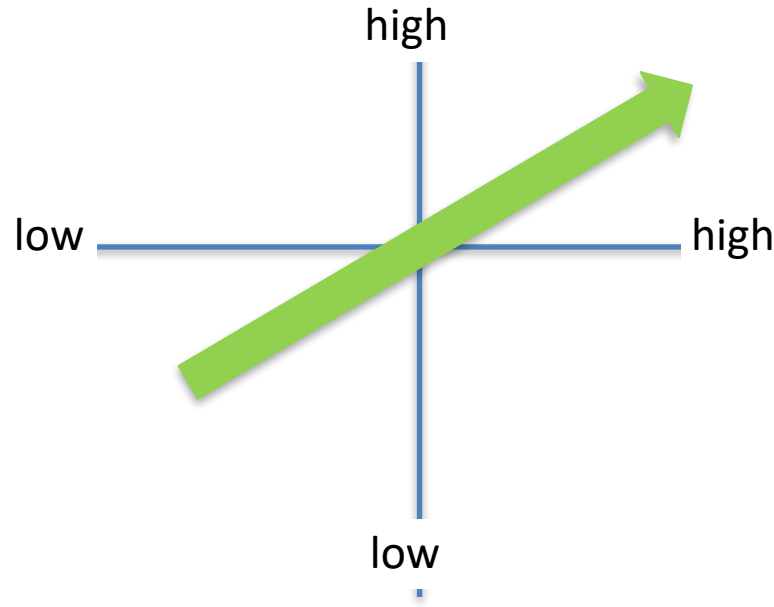
Published 11 September 2026 · DOI [10.5281/zenodo.22737750](https://doi.org/10.5281/zenodo.22737750) ▶ English

Over the last few months, the mathematical capabilities of LLMs have improved dramatically, to the point that they can solve major outstanding problems in many fields of mathematics. However, the push by AI companies to solve mathematical problems as a benchmark is detrimental to the science of mathematics, and to the mathematical community. The goals of the AI companies and the goals of the mathematical community are severely misaligned. We see these as part of broader alignment issues impacting other scientific and creative professions, as well as the whole of society.

Quadrants of competencies

GAI proficiency

required to remain competitive and reap the benefits of this technology



Domain

Expertise

required for asking the right questions and for validation & verification



Theoretical capability and observed usage by occupational category

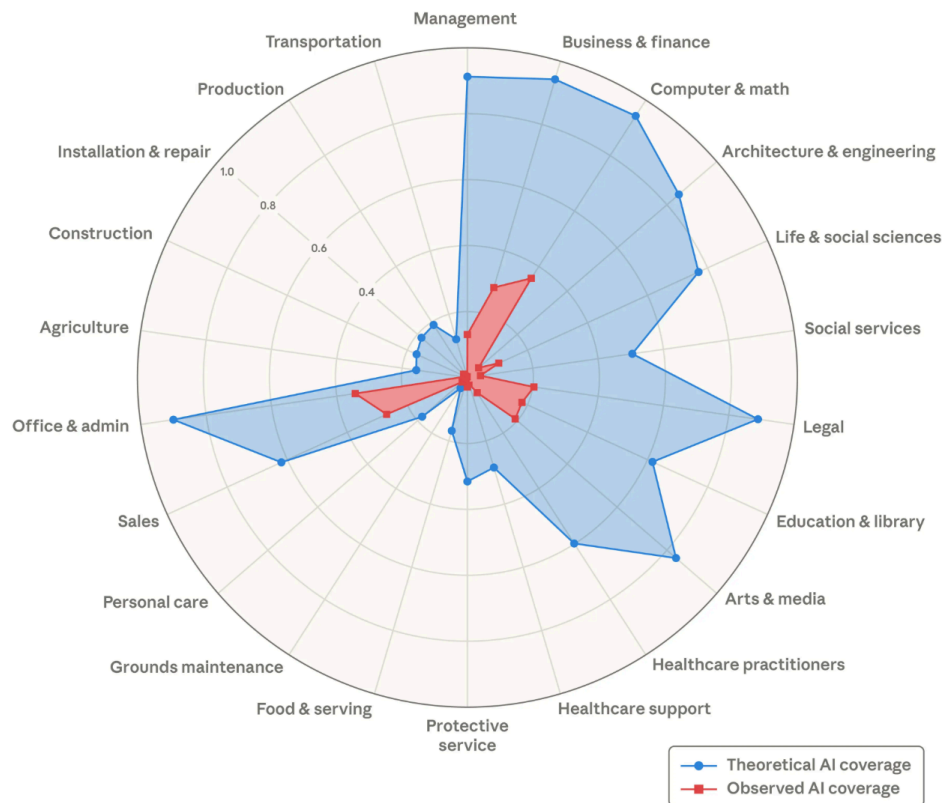
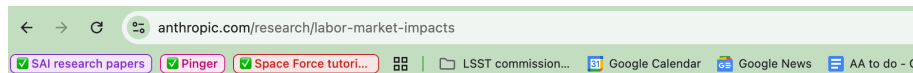


Figure 2: Theoretical capability and observed exposure by occupational category
Share of job tasks that LLMs could theoretically perform (blue area) and our own job coverage measure derived from usage data (red area).

March 2026 Anthropic report



Labor market impacts of AI: A new measure and early evidence

Mar 5, 2026

I was told Anthropic's programmer productivity is 8X higher when using Claude.

Ethical, Professional, and Environmental Aspects

- Act responsibly: verify and validate AI output
- Strive for transparency and clarity in AI use.
- Reduce your carbon impact- use the minimal model that can get the job done.
- Keep abreast of rapidly changing AI use norms and expectations
 - <https://journals.aas.org/author-llm-guidelines>
- Disclose, disclose, disclose!



Norms of Use, Transparency, and Disclosure

- Student-advisor expectations
- Collaboration expectations
- Institutional expectations
- Agency expectations
- Journal expectations
- ...



Reciprocal expectations for AI use. V 1.0

This checklist is intended to prompt a discussion about the use of generative AI in a PhD student's course of study. The thesis advisory committee, or its equivalent, should confirm in writing that a shared understanding between student and advisor has been achieved. Check the yes or no box as appropriate, reflecting shared expectations regarding potential uses of generative AI. All use of AI should be appropriately acknowledged.

	yes	no
Preparing a reading list of relevant books and papers, for research topics		
Preparing summaries of background readings and papers		
Brainstorming assistance		
Generation of outlines for presentations, papers, and dissertation		
Generation and revision of rough drafts and other research materials		
Polishing and critique of human-drafted materials		
Generation and revision of slide decks for presentations		
Generation and revision of graphics and videos		
Generation and debugging of computer code		
Delegation of entire research questions to agentic AI tools		
Assistance in revising and polishing narrative material and presentations		
Preparing job application materials.		
Drafting letters of recommendation		
Student submission of papers and dissertation chapters to AI for feedback		
Advisor submission of papers and dissertation chapters to AI for feedback		
Other:		

Student: _____

Advisor: _____ TAC committee chair: _____

Date: _____

Outline

Chat interface examples

damped harmonic oscillator

knowledge extraction from a data file.

Image generation and multimodal AI- images, audio, podcasts, video...

Google Notebook tool- constrained output with reach-back into source documents

Beyond the chatbot:

Code generation and debugging- AI integrated into notebooks: Colab, Jupyter

Agentic AI – unleashing the beast on your own machine: Claude Code, Command Line Interface

Other aspects, not covered today:

System administration on your laptop.

Integration with other tools- Overleaf, Excel, PPT...



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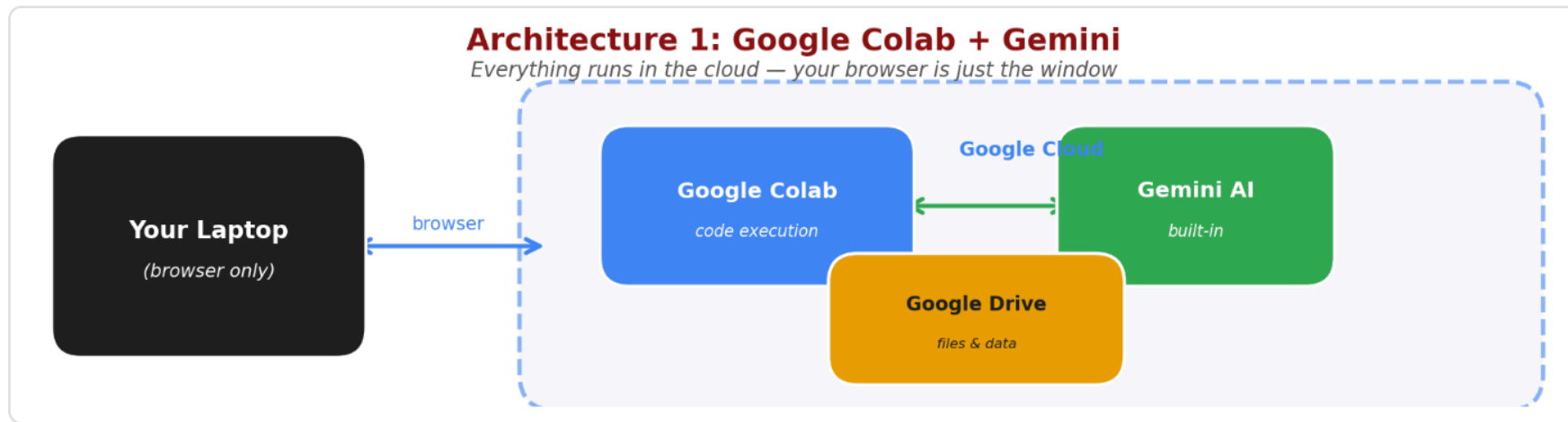


Code generation and debugging

- You can cut and paste error messages into GPT, Claude, Gemini and then copy suggested code revisions, and edit source code. Cumbersome!
- Python notebooks can have direct AI integration, gives faster iterations compared to cutting and pasting error messages into chat interface.
- Allows for natural language modifications, and explanation of code.
- Options:
 - 1) Colab notebook in Google drive folder, w/ integration of Google Gemini. Notebook runs on Google cloud hardware, no persistent environment, need to pip install packages each time notebook runs. There is a way to mount your Google drive for data access.
 - 2) Enable AI assistant within a Jupyter notebook, requires account authentication via API key or account validation. The notebook could run on your own laptop.
 - 3) Use Claude Code (or the equivalent) as the main user interface, and have it modify code that runs locally on your machine, either as notebook or Python source code.



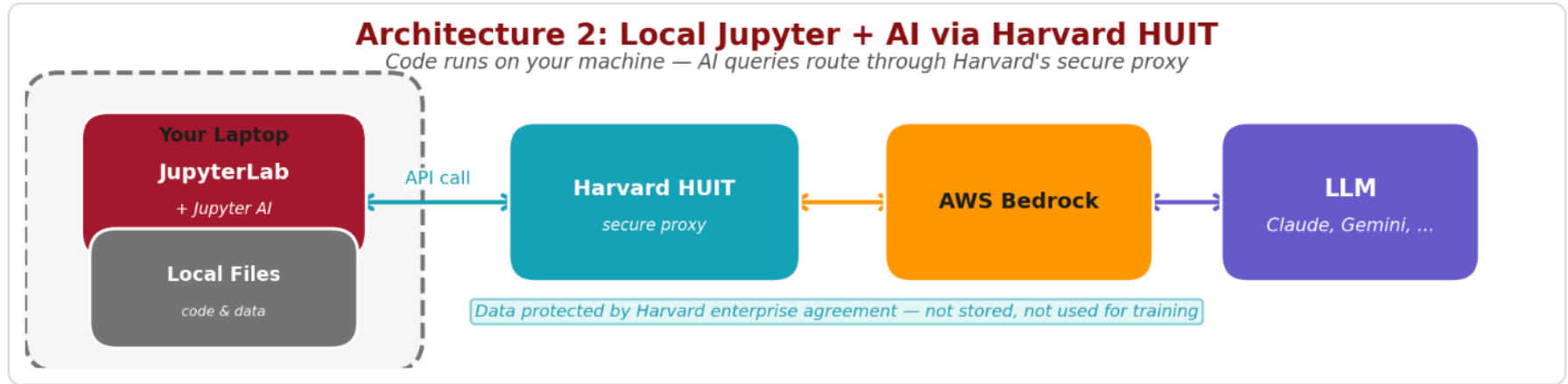
Architecture 1 of 3: Laptop as thin client, data and execution in the cloud



Architecture 2 of 3: No AI interface, data and execution are local, but AI calls can happen from within Python code, via API calls to AI system.

Architecture 2: Local Jupyter + AI via Harvard HUIT

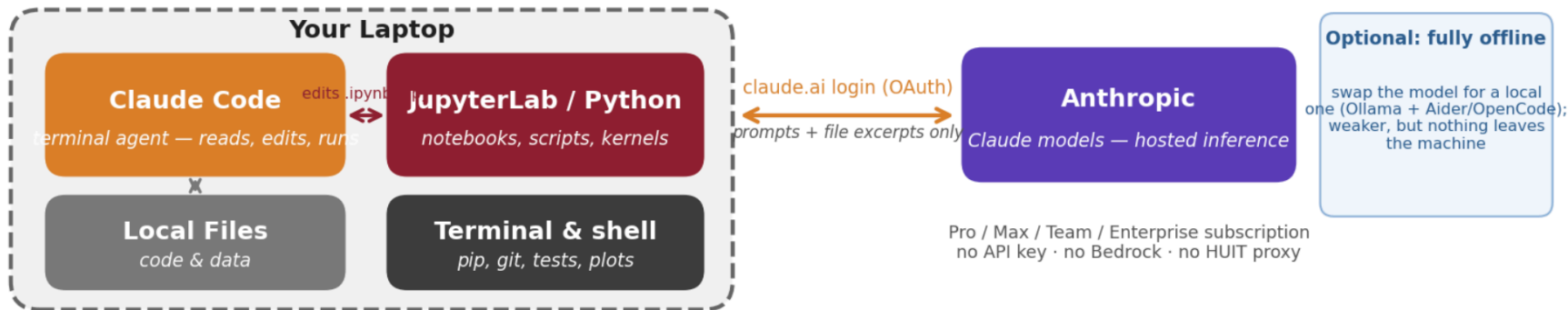
Code runs on your machine — AI queries route through Harvard's secure proxy



Architecture 3 of 3: Local data and execution, AI calls are external but through locally running AI system that has access to your computer

Architecture 3: Local Jupyter / Python + Claude Code

Code, files and the agent all run on your machine — only the model call leaves the laptop



Magic Wand in Colab notebooks

7. type: Explain this code line by line

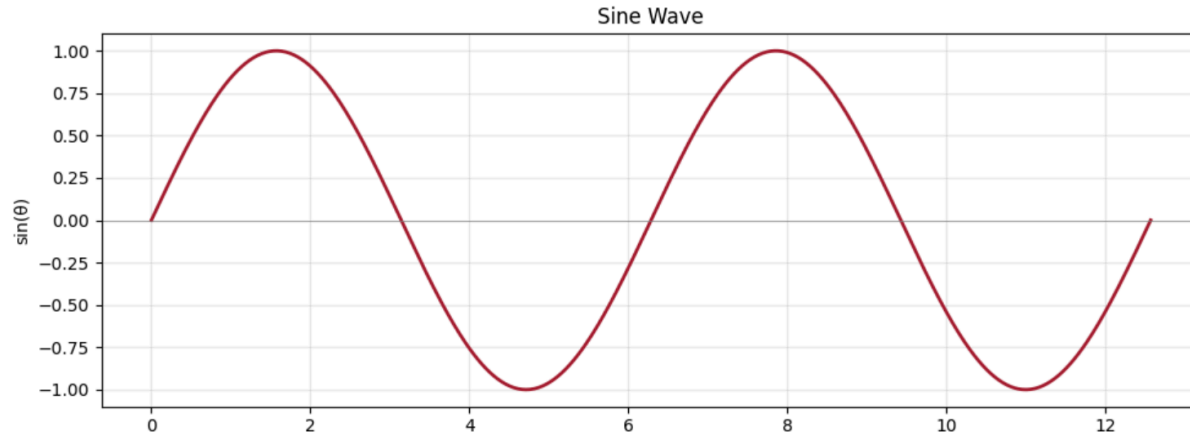
5. Read Gemini's explanation — does it match your understanding?



```
11 ▶ import numpy as np
12 import matplotlib.pyplot as plt

theta = np.linspace(0, 4 * np.pi, 500)
y = np.sin(theta)

plt.figure(figsize=(10, 4))
plt.plot(theta, y, color='#A51C30', linewidth=2)
plt.xlabel('θ (radians)')
plt.ylabel('sin(θ)')
plt.title('Sine Wave')
plt.axhline(y=0, color='gray', linewidth=0.5)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



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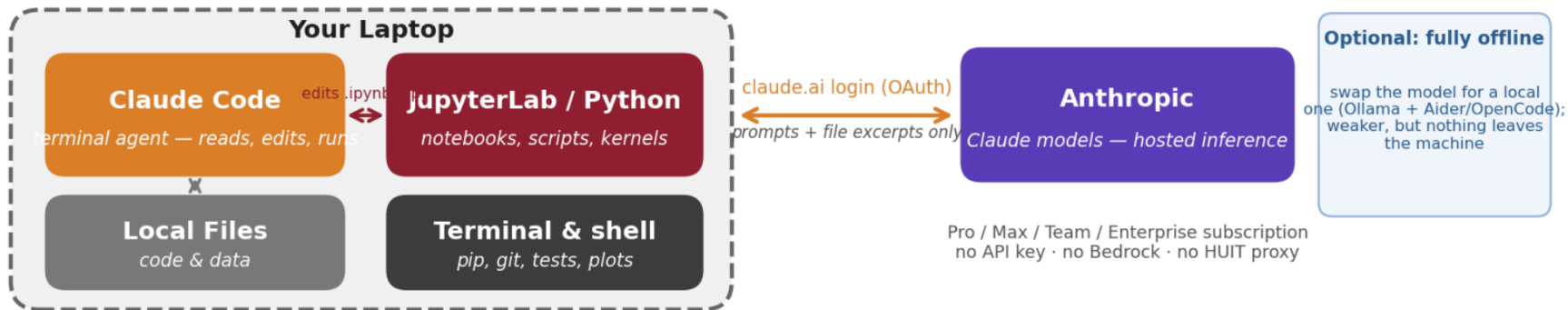
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Folders and files
Model selection
CLI vs. Claude GUI
Skills and / commands



Claude Code: the slash commands you'll actually use

Type `/` at the prompt to see the full list; commands take effect immediately in the current session.

● Session & context

`/clear`

Start fresh — wipe the conversation history

`/compact`

Summarize the context to free up room (add focus hints)

`/context`

Show what's filling the context window

`/resume`

Pick up a previous conversation

`/rewind`

Roll back to an earlier checkpoint

`/export`

Save the transcript to a file

● Model & cost

`/model`

Switch models (Opus, Sonnet, Haiku...) or effort level

`/fast`

Toggle fast mode for quicker output

`/usage`

Plan limits and how much you've used

`/cost`

Token counts and dollar cost of this session

`/status`

Version, model, account, and connection info

`/doctor`

Diagnose installation and setup problems

● Project setup

`/init`

Generate a CLAUDE.md describing this codebase

`/memory`

Edit persistent memory / CLAUDE.md files

`/config`

Theme, notifications, and other settings

`/permissions`

Manage which tools run without asking

`/mcp`

Connect and inspect MCP servers

`/hooks`

Run shell commands on tool events

● Working & help

`/help`

Command reference and keyboard shortcuts

`/agents`

Create and manage custom subagents

`/review`

Request a code review of your changes

`/plan`

Enter plan mode — design before editing

`/vim`

Toggle vim-style editing in the prompt

`/bug`

Report a problem to Anthropic

Bonus: `!` `cmd` runs a shell command inline · `#` `note` saves a fact to memory · `Shift+Tab` cycles permission modes

Limitations

- Collaborative tools (N of us, M agents) not well developed.
- Validation and verification of code is a challenge.
- AI tools and names change rapidly, need to keep abreast of it.
- Cybersecurity and research integrity
- Having an AI system layered on top of your computer is, uh, unsettling. There are instances of them doing unhelpful things.



But things are changing, and changing fast...



2025-9-9

An AI system to help scientists write expert-level empirical software

Eser Aygün^{1,*}, Anastasiya Belyaeva^{2,*}, Gheorghe Comanici^{1,*}, Marc Coram^{2,*}, Hao Cui^{2,*}, Jake Garrison^{3,*},
Renee Johnston^{2,*}, Anton Kast^{2,*}, Cory Y. McLean^{2,*}, Peter Norgaard^{2,*}, Zahra Shamsi^{2,*}, David Smalling^{1,*},
James Thompson^{2,*}, Subhashini Venugopalan^{2,*}, Brian P. Williams^{2,*}, Chujun He^{2,4,**}, Sarah Martinson^{2,5,**},
Martyna Plomecka^{2,6,**}, Lai Wei², Yuchen Zhou², Qian-Ze Zhu^{2,5,**}, Matthew Abraham², Erica Brand², Anna
Bulanova¹, Jeffrey A. Cardille^{2,7}, Chris Co², Scott Ellsworth², Grace Joseph², Malcolm Kane², Ryan
Krueger^{2,5,**}, Johan Kartiwa², Dan Liebling², Jan-Matthis Lueckmann², Paul Raccuglia², Xuefei (Julie)
Wang^{2,8,**}, Katherine Chou², James Manyika², Yossi Matias², John C. Platt², Lizzie Dorfman², Shibl Mourad^{1,‡}
and Michael P. Brenner^{2,5,‡}

¹Google DeepMind, ²Google Research, ³Google Platforms and Devices, ⁴Massachusetts Institute of Technology, ⁵School of Engineering and Applied Sciences, Harvard University, ⁶Google Cloud, ⁷Faculty of Agricultural and Environmental Sciences, McGill University, ⁸California Institute of Technology





Paper Title and Authors:

Submit to ApJ

Submit to AJ

Submit to Nature

That's it!

Have fun.

